

Simulating solid-state battery cathode manufacturing via wet-processing with resolved active material geometries

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ABSTRACT

Prior to the development of a solid-state battery cell, researchers have limited knowledge about the microstructure of the electrodes and how they are affected by manufacturing. Therefore, numerical simulations can be considered as a powerful tool to link the fabrication process to the final microstructure of the electrode. In this paper, a numerical simulation of a wet-processed solid-state battery cathode with a formulation of 75 % $\text{LiNi}_0.5\text{Mn}_{0.5}\text{Co}_{0.5}\text{O}_2$ (NMC), 17.5 % LPSCl, 5 % Timcal C65 and 2.5 % Polyisobutene (PIB) is presented. From nano-computed tomography images, realistic shapes of active material particles are extracted and used in the simulation, which is well-calibrated to experimental data. In particular, we study the effects of calendaring on the microstructure of the simulated cathode and deduce structure-property relations.

1. Introduction

Lithium-ion (Li-ion) batteries are considered a cornerstone of electrochemical energy storage technologies. Ever since their commercialization by Sony in 1991, research has consistently pushed the state of the art of these systems [1,2]. Over the years, researchers have shifted some of their focus from solid-state physics and chemistry to understanding the architecture and texture of the electrodes [3]. Such a strategic shift is necessitated by consumer demand for safer and more energy dense batteries at a lower cost. Particularly, safety concerns such as the flammability of Li-ion batteries [4] and further degradation effects are major factors that can discourage their use in automotive technologies and thus limit their widespread adoption [5]. Advanced high-performance battery chemistries, that mitigate the inherent shortcomings of traditional Li-ion batteries, as a successor to the current state of the art is therefore important [6]. Solid-state batteries that replace the liquid electrolyte for ionic conductivity by a solid electrolyte (SE), are considered an important candidate for this, as they allow for higher theoretical energy densities.

One of the challenges impeding the widespread commercialization of

solid-state batteries is the complexity arising from their electrodes manufacturing. Such manufacturing processes should ensure optimal interfaces between the active material and solid electrolyte to produce electrodes with good electronic and ionic conductivities. Optimized solid-state battery manufacturing processes are of paramount importance to allow for a low cost adaptation. Different types of battery manufacturing technologies are currently under development to achieve this in a cost-effective manner. On the laboratory-scale, cathode materials need to be laboriously ground and mixed [7]. An emerging alternative is a solvent-free manufacturing approach, in which, a molten binder, active material and solid electrolyte are extruded at high temperatures [8]. Such process have already been successfully digitalized by us recently by using a discrete element method approach [9]. Extrusion, however, is a technology that is not yet mature enough for electrode mass production.

Roll-to-roll methods, such as wet processing, are already wide-spread for the manufacturing of Li-ion batteries [10]. In wet processing, a slurry, containing the electrode components and a solvent, is tape-casted onto a current collector (CC) and dried. Once the solvent has completely evaporated, the resulting electrode is calendared to a desired thickness

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[11]. These production methods have already been optimized for Lithium-ion batteries and benefit from economies of scale for low-cost production [12]. Similar processes can be used for solid-state battery manufacturing. However, replacing the liquid electrolyte with a SE introduces other disadvantages in terms of outcome performance. Particularly, since the electrolyte is no longer able to wet the active material efficiently, interfaces are created between different materials, which induces capacity fade and mechanical instabilities thus reducing ionic conductivity on account of sluggish interfacial kinetics [13,14]. Understanding the processability of solid electrolytes is also essential to develop optimized electrode microstructure and understanding the interfacial evolution during processing and operation is important to maintain contact at the interface and thus ensure a continued ionic conductivity [15].

In this regard, understanding the microstructural evolution of electrodes during manufacturing and the impact of manufacturing parameters on the electrochemical performance is critical [3,16,17]. Researchers often use non-destructive means, such as computer-generated tomography imaging to understand the internal microstructure [18]. This approach is time consuming and is often unable to accurately distinguish all the material components. Most segmentation methods involve separating structures by their gray scale in the nano-Computer Tomography (CT) image. If similar gray scales are present, different phases cannot be uniquely determined by segmentation processes [19,20], which adds difficulty to the analysis of microstructures. Regarding this, computational methods can be used to gain insights into the electrode microstructure. Historically, microstructures were simply randomly generated [21] and in fact, most computational electrochemical studies are performed on stochastically generated structures [22]. Particles corresponding to active material (AM), SE, and mixtures of binder and carbon black (Carbon-Binder or CB), were randomly placed into a predefined volume to numerically study the impact of, for example, particle distributions on the electrochemical performance. However, experimentally, on both the laboratory and industrial scales the method of producing the battery electrode influences the end-result, morphologically and thus electrochemically [18,23]. Coupling the microstructure to the manufacturing parameters enables a more complete approach to understand the electrochemistry. In response to lab and industry demand, computational models were developed to study the microstructure [24] and by our ERC-funded ARTISTIC project to link the manufacturing process to the electrochemical performance of batteries [25–27]. In the ARTISTIC models, a combination of physics based and machine learning models were developed as a digital twin of the manufacturing process chain. These models began with the simulation of Nickel Manganese Cobalt oxide (NMC)-based cathodes for lithium-ion batteries [25] and were later expanded to include other battery chemistries such as all solid-state batteries [16]. Such models generally consist of performing a coarse-grained molecular dynamics or discrete element method simulation of each step in the electrode processing line, followed by Lattice Boltzmann Method and Finite Element Method simulations of the electrolyte filling [28] and the resulting electrochemical behavior. A slurry is generated by mixing the electrode constituents, then dried by solvent removal and finally calendered. A refinement of the model was later performed by our group to account for microstructural evolution at different drying rates [29]. The force fields describing particle interaction were adapted in order to account for capillary forces between particles. A continuum approach was later introduced by Zihur et al. by implicitly modeling capillary, buoyancy, friction forces and binder migration as well as the effects of drying on layer thickness through the convection-diffusion equation [30]. A pure continuum approach to electrode drying has also been proposed by Susarla [31] and Nikpour [32]. Recently, coupled fluid dynamics and discrete element method simulations of fluid-structure interactions have emerged to simulate a coarse model of the drying process [33]. These models use a volume of fluid approach with a volumetric source term to study the multiphase

nature of solvent evaporation, however they do not consider the distribution of the inactive binder phase. Approaches to calendaring simulations have previously been developed [34–36] and recently further extended by our group to involve deformation [37] and fracture of realistic particle morphologies upon electrode calendaring [38]. For solid-state battery manufacturing, computational generation of electrode microstructures is often limited to the stochastic generation approach [39]. This is a computationally economical approach to study the influence of the microstructure on the electrochemical properties, which does not take into account the complex effect of manufacturing on the microstructure. Alabdali et al. [16], from our group, developed the first model linking the microstructure of a solid-state battery to its manufacturing process, by taking into account spherical shapes for the active material.

As a proof-of-concept, a further development of the existing simulation method of the solid-state battery manufacturing line with wet processing, developed by Alabdali et al. [16] is reported in this article. The novelty of the work reported here lies in the extraction of realistic AM particle shapes from nano-CT images and using these in the solid-state battery electrode manufacturing simulation workflow, unlike in previous work from our group, in which purely spherical particles were used [22,40]. Previously, the inclusion of realistic shapes was only done for Li-ion batteries [37]. This addition in the simulations allows for a more accurate determination of electrochemical properties of the solid-state battery electrode. Including such realistic shapes allows for scientists to improve their understanding of the actual paths electrons and ions are able to take through the microstructure and allows for the determination of more realistic interface properties. To the best of our knowledge, computational workflows considering realistic particle shapes have never been considered for solid-state batteries. As a proof-of-concept, our aim is to present an approach with existing data that works with limited computational resources. Limitations and possible extensions of our model are detailed.

This article begins by introducing the theory and methodology. Here, the simulation framework is described and the process used to represent realistic shapes is explained. Subsequently, attained results are analyzed and discussed. Finally, conclusions are drawn and perspectives of our work are presented.

2. Theory and methodology

The description of the methodology and theory behind the employed framework ensues. We begin by describing the experimental process through which the studied electrode is prepared. Next, the method for generating AM particles with realistic shapes is described and finally the discrete element model (DEM) used in the manufacturing simulation is presented.

2.1. Slurry preparation

The SE cathode slurry, that is composed of 75 % $\text{LiNi}_0.9\text{Mn}_{0.5}\text{Co}_{0.5}\text{O}_2$ (NMC), 17.5 % LPSCl, 5 % Timcal C65 and 2.5 % Polyisobutene (PIB), is prepared in order to characterize its density. Each material constituent was dried under vacuum at 120 °C and then stored in a glovebox before use. To prepare the slurry, first NMC was hand-mixed alone in a mortar for 5 min, then the Timcal C65 was added to the mortar and ground by hand for 10 min. Subsequently, the SE LPSCl was added and all powders were ground together for another 10 min. The solution of 9 % polyisobutylene (PIB) in p-xylene was added into a vial with the magnet stirrer. Afterwards, the resulting powder was added and mixed with 2 g p-xylene to reach a solid content of 50 %. Finally, the slurry density was measured in an Anton Paar densimeter with the reference DMA 4100 M, right after being prepared.

2.2. Tomographic imaging and particle extraction

Nano-Computed Tomography (nano-CT) images obtained from an electrode containing $\text{LiNi}_{0.6}\text{Mn}_{0.2}\text{Co}_{0.2}\text{O}_2$ are used to extract realistic particle shapes of the AM. Such images provide high-resolution data of the AM skeleton and allow the visualization of pores in the micro-structure, which is an important aspect to consider for our modeling framework validation.

Note that this is not the same material as used in the slurry formulation described previously. We chose these nano-CT images on account of the data availability, which is sufficient for the proof-of-concept work reported in this article. A description of the methodology used to extract particle morphologies briefly ensues, it is based on a similar methodology employed for previous models developed in our group [37].

The raw three-dimensional nano-CT images are fed into ImageJ [41, 42]. An example of a two-dimensional slice of the nano-CT image is shown in Fig. 1. Different gray scale values in the image correspond to different material phases. A Weka segmentation plugin [43] is trained to classify each pixel according to its gray scale into its corresponding phase. A Fast Random Forest classifier is used for classification. Carbon additives, binder and solid electrolyte were indistinguishable to nano-CT imaging from one another, which limits the use of realistic SE particle shapes in our simulations. Therefore, in total, three different phases were identified. They correspond to an AM phase, a pore phase and a phase consisting of SE, carbon black and binder. After the segmentation is completed, the AM particles are extracted. A schematic representing this process is shown in Fig. 2. On account of the segmentation, high-resolution images of the AM can be extracted from the nano-CT data and used in subsequent simulations. As a proof-of-concept, and with the purpose of demonstrating the methodology introduced in the present article, the high-resolution particles were not used in the final simulations. Due to the high computational cost of simulating the high-resolution AM, a pixel position averaging technique was employed. A sliding-algorithm traversed the high-resolution three-dimensional nano-CT images. Sets of three adjacent pixels were averaged after sliding in each direction one-by-one. Whenever the average value surpassed a

threshold value of 0.5, it was taken as the center of an element of the realistic particle. Such particles will be denoted as primary particles and can be thought of as a monocrystalline structure within the poly-crystalline NMC. The methodology employed resulted in coarser shapes, represented in Fig. 2. A higher resolution can be employed in the simulation if it is desired.

Ultimately, the realistic shapes of the AM particles where approximated by the use of a multisphere method, in which particle shapes are approximated by a number of overlapping spheres, whose centers are fixed relative to each other, by the use of harmonic bonds. Hereby, computational cost is reduced by the use of this approximation. The watershed algorithm [41,42] is employed to separate pixels connecting AM particles. Resulting images are then post-processed to search for AM particles, which will be denoted secondary particles in this article. Secondary particles with sizes ranged into the targeted particle size distribution are extracted for the subsequent simulation.

2.3. Discrete element method

The DEM workflow used to simulate the manufacturing process is based on the standard wet-processing protocol used in a manufacturing pilot-line. A slurry is created, dried to form an electrode and then calendered. In Fig. 4, snapshots of the simulations of the wet-processing are shown to illustrate the procedure.

In our simulation framework, the slurry generation implies generating particles at random positions inside a large simulation box with periodic boundary conditions in all directions as an initial configuration. SE and the CB-solvent domain are modeled as spheres, whilst the AM shape is resolved according to the methodology described in Fig. 2 and based on [37], shown in Fig. 3. One assumption of the model is that the carbon and binder are both thoroughly mixed and thereby homogeneously distributed within the Carbon Binder Domain (CBD). Such an assumption is typically made while analyzing CBD morphology. [44] All simulations are performed using the open source molecular dynamics code LAMMPS [45]. The motion of a particle i , with mass m_i , that is located in the three dimensional vector space $\vec{r}_i$ is described according to the Langevin equation with an external force contribution, as well as three additional rotational degrees of freedom. Rotational motion is calculated based on the following equation of motion

$$I_i \frac{d\omega_i}{dt} = M_i. \quad (2.1)$$

Here, I_i is the moment of inertia, ω_i is the angular velocity and M_i is the resulting contact torque acting on a particle i . Translational motion is calculated by the Langevin equation with an additional force contribution, it is written as

$$m_i \frac{d^2 \vec{r}_i}{dt^2} = -\lambda \frac{d\vec{r}_i}{dt} + \vec{f}_i^{\text{ext}} + \vec{\eta}(t). \quad (2.2)$$

Eq. (2.2) describes the force acting on a particle as a combination of a velocity-dependent friction term, a stochastic term and external forces. The damping coefficient, λ acts against the direction of motion of the particle. A time-dependent stochastic force term, $\vec{\eta}(t)$, with a Gaussian probability distribution, is included to describe the Brownian motion of the particles. An external force $\vec{f}_i^{\text{ext}}$ is used to describe the explicit interaction between each particle within a cut-off radius $|\vec{r}_c|$ and is calculated as the negative spacial gradient of the potential energy, it is given as,

$$\begin{aligned} \vec{f}_i^{\text{ext}} = & -\vec{\nabla} U_{\text{pot}} = -\vec{\nabla} U_{\text{LJ}}(\vec{r}_i, \vec{r}_j) \\ & + \vec{F}_{\text{JKR}}(\vec{r}_i, \vec{r}_j) - \vec{\nabla} U_{\text{harm}}(\vec{r}_i, \vec{r}_j). \end{aligned} \quad (2.3)$$

Explicit interactions are modeled as a combination of the Lennard-Jones

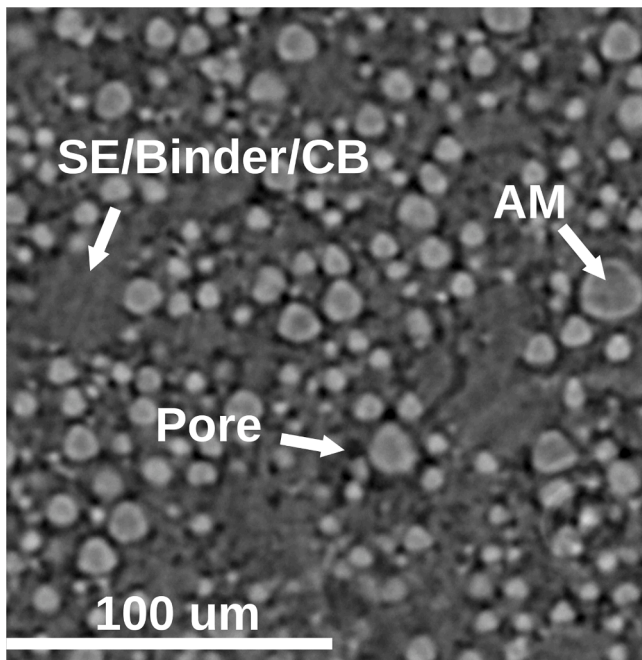


Fig. 1. Slice of a tomography image in which the different phases are resolved according to their gray scale. Almost black pixels correspond to pores, while the dark gray pixels are indicative of the SE/binder/CB agglomerates. The lighter, circular shaped objects represent the active material particles.

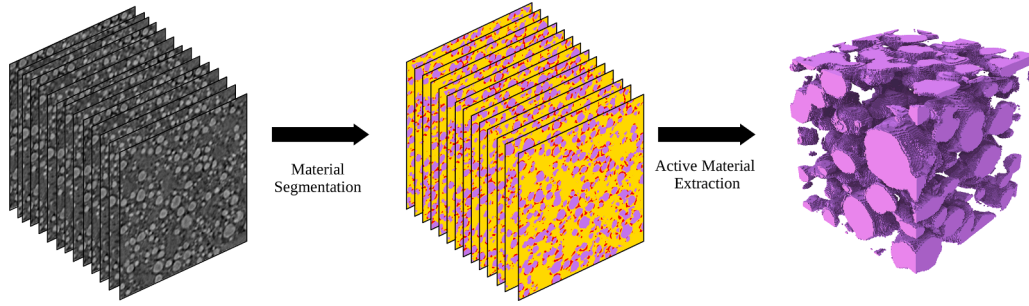


Fig. 2. Schematic representation of the process used to extract the particle shapes from tomography images. Slices of tomography images (left), in which the different phases are resolved according to their gray scale value are segmented into the different material phases. A Fast Random Forest classification is used to assign each voxel into their different phases (center). Here, the purple phase represents the AM, red are pores and the yellow phase corresponds to a mixture of binder, carbon black and SE. The set of classified tomography images is used to extract AM particle skeleton from the tomography data, represented on the right.

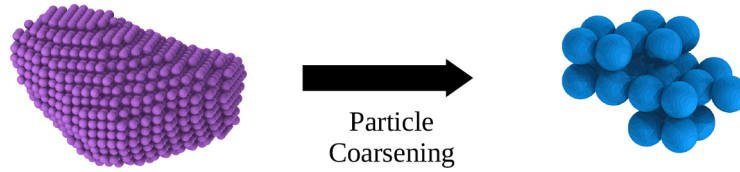


Fig. 3. An illustration of how the high resolution particles look before (left snapshot) and after (right snapshot) the coarsening procedure described in the text.

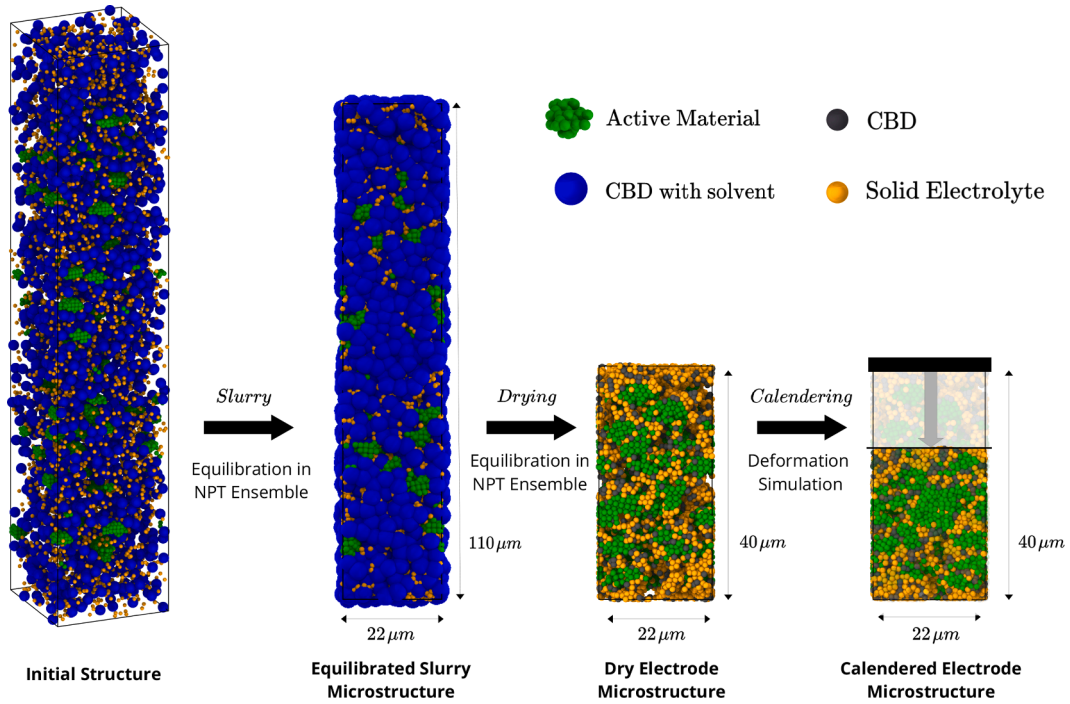


Fig. 4. Snapshots of each simulation step illustrating the manufacturing process. First, an image showing the initial structure is shown. Once this structure is equilibrated in the isochoric-isobaric ensemble (NPT), we get the equilibrated slurry microstructure. The corresponding final microstructure is dried by shrinking the CBD with solvent to simulate the solvent removal and once again equilibrated in NPT to attain a dry electrode microstructure. The left image shows the slurry after its initial equilibration. Bright blue spheres illustrate the CB-solvent domain. AM and SE are obscured by the CB domain. An electrode after solvent removal is shown in the center. Yellow spheres correspond to the solid-electrolyte particles, green particles are the realistically shaped AM and dark blue depicts the CB domain after the solvent is removed. On the right-hand side one can see a calendered electrode. Videos showing the evolution of the simulations, highlighting springback after calendering, are included in the Supplementary Material.

(LJ) force, a Johnson–Kendall–Roberts (JKR) contact force [46] and harmonic bonds, the parametrization of which is described in the subsequent section.

The LJ potential marks the first contribution to the external forces in Eq. (2.3) is described as a function of distance r_{ij} between the centers of two particle i and j ,

$$U_{LJ}(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right] \quad \text{if } r_{ij} < r_c, \quad (2.4)$$

if r_{ij} is within a cutoff distance r_c , otherwise the potential nullifies. The variable ϵ describes the depth of the potential well and is a measure of

interaction strength between two particles, while the value σ indicates the distance at which the potential energy is zero and is indicative of how close two particles can get. The LJ potential can be thought of as describing a repulsive force acting at small distances and van-der-Waals forces at larger distances. LJ Forces acting between different particles are calculated according to empirical mixing laws previously used in Alabdali et al. [16], Xu et al. [37] and based on [47]. More precisely, the value for ε used to determine the force is based on the geometric mean,

$$\varepsilon = \sqrt{\varepsilon_i \varepsilon_j}, \quad (2.5)$$

and the value for σ is calculated based on the arithmetic mean,

$$\sigma = \frac{\sigma_i + \sigma_j}{2}. \quad (2.6)$$

Here, the values ε_{ij} and σ_{ij} are the values from the interaction of a single particle with another particle of the same type.

A Johnson-Kendall-Roberts (JKR) model [46] is employed to simulate the mechanical interaction and frictional force of granular particles, it is described as,

$$\vec{F}_{\text{JKR}} = \left(\frac{4Ea^3}{3R} - 2\pi a^2 \sqrt{\frac{4\gamma E}{\pi a}} \right) \vec{n}. \quad (2.7)$$

In Eq. (2.7), R denotes the effective radius $R = R_i R_j / (R_i + R_j)$, in which $R_{i/j}$ denotes the radius of particle i/j . Furthermore, $\vec{F}_{\text{JKR}}$ is parameterized by the effective Young's modulus,

$$E = \left(\frac{1 - \nu_i^2}{E_i} + \frac{1 - \nu_j^2}{E_j} \right)^{-1} \quad (2.8)$$

and a surface tension energy density γ . The contact zone radius a is calculated through the overlap distance,

$$\delta = \frac{a^2}{R} - 2\sqrt{\frac{\pi\gamma a}{E}}, \quad (2.9)$$

between two particles and only acts if $\delta > 0$. $\vec{n}$ is the unit vector connecting the centers of these particles.

Finally, for primary particles belonging to the same secondary AM particle, harmonic bonds are included between nearest and next-to nearest neighbors, these are an artificial contribution that enable the realistic particles to retain their shape and remain rigid. The harmonic potential is described as

$$U_{\text{harm}}(r_{ij}) = \sum_{i=1}^N \frac{\kappa}{2} (r_{ij} - r_0)^2. \quad (2.10)$$

Harmonic forces are described as the sum over all harmonic interactions between nearest and next-to nearest neighbors. They only act between primary particles belonging to the same secondary AM particle and on account of an artificially high bond strengths allow only for minimal deformation. The harmonic bond coefficient κ is indicative of the strength of the interaction and r_0 is the harmonic bond length, which describes the distance at which two particles are in the potential minimum.

For each step of the manufacturing process, namely slurry development, slurry drying and electrode calendering, there exists one simulation method, described in the current section. Firstly, a slurry development simulation is performed. Here, particles are stochastically placed into a large simulation box with periodic boundary conditions in all directions in accordance with a given experimental formulation. An equilibration simulation is performed in the isochoric-isobaric ensemble by coupling it to a heat-bath for constant temperature and a barostat for constant pressure simulations. Once a constant temperature and pressure is attained, the structure is further allowed to evolve until a time-stopping condition is reached. The final particle positions of the slurry

generation simulation are then included in the subsequent drying step of the processing line. The drying step consists of taking the final structure obtained from slurry equilibration, in which the CB particles have a diameter $d_{\text{CB}}^{\text{slurry}}$ and replacing the CB particle diameter with that of dry CB particles $d_{\text{CB}}^{\text{electrode}}$ with the volume of the solvent removed,

$$d_{\text{CB}}^{\text{slurry}} \rightarrow d_{\text{CB}}^{\text{electrode}}. \quad (2.11)$$

Such a method corresponds to a drying model which is intended to describe experiments conducted at low drying rates [25]. Models to simulate structure formation at high drying rates also exist (previously reported by us) [29], however they are out of scope for this article. The dried structure is equilibrated in the isochoric-isobaric ensemble. The force field parameters between all particles are recalibrated in order to attain experimental porosities.

Finally, a calendering simulation [36] is performed on the dried electrode microstructure. In contrast to modeling the CBD as an implicit bond [34], here both AM and CBD are considered explicitly. Calendering is a crucial process in solid-state cathode manufacturing, as it enhances the contact between each material constituent as well as increases the energy density and decreases the ionic tortuosity of the system. It is simulated by an introduction of a calendering roll on the top part of the microstructure along with the CC on the bottom part of the microstructure. If an active material particle is placed over the CC in the periodic boundary condition at the end of the drying stage, it is simply cut at that point in the calendering simulation. Both the calendering roll as well as the CC interact with the electrode microstructure through the JKR-potential. The calendering roll is lowered at a constant velocity until the desired calendering degree is attained. Afterwards it is removed at the same velocity to allow for springback phenomena to occur, as observed experimentally.

2.4. Force field calibration

Our approach aims to be chemistry neutral, potentially describing electrode manufacturing for a wide variety of chemistries [16,36,48]. Pair-interactions, according to Eq. (2.3), between different material constituents are calibrated in such a way that the properties of the simulated structure mimics the properties of the experiments. All forces are calibrated in the slurry equilibration and the drying step. In particular, during the equilibration of the slurry, the simulated model should display the same density as the experiment. Typically such simulations also require a calibration of the viscosity [16], however, similarly as in our previous work on Li-ion batteries [37], on account of the large computational cost of the realistically shaped particles, this was not considered in this proof-of-concept work. Once both the experimental and simulated densities show an agreement, the model is considered to be well-calibrated and the resulting microstructure may be used in the subsequent processing steps. In this work, a multi-objective Bayesian optimization algorithm [49] is employed to calibrate the model. The optimization model transforms a multi-objective problem, i.e., the optimization of several target parameters β from a set $\mathcal{N}$, into a single cost-objective function,

$$C = \sum_{i \in \mathcal{I}} \alpha_i \cdot (1 - \beta_i)^2, \quad \text{with } \sum_{i \in \mathcal{I}} \alpha_i = 1 \quad (2.12)$$

In which $\alpha_i \in [0, 1]$ factors in the weights, i.e., the relative importance of each contributor to the cost function. Our proof-of-concept model only optimizes the density, reducing the problem to $\beta_i = \rho_{\text{sim}}/\rho_{\text{exp}}$. The Bayesian optimization algorithm calculates a posterior distribution over the existing data set and iteratively proposes new force field parameters in such a way to attain a minimization of the cost function, $\text{argmin}(C)$, and thereby reach a local minima. As long as a minimum has not been reached, after each iteration the new set of force field parameters is used to restart a simulation and hereby create a new data set, the results of

which are then reiterated and passed in the cost function.

2.5. Determination of ionic and electronic conductivities

The method for the determination of ionic and electronic conductivities has been described in detail in a previous publication. These were performed employing the DiffuDict and ConductoDict modules of GeoDict (Math2Market) [16]. Ionic conductivity is calculated from the diffusive flux across the electrode, obtained from a steady state diffusion simulation in which a difference of concentration of 1 mM is fixed between opposite planes of the simulation box. Then, the effective diffusivity is calculated as the absolute value of the flux multiplied by the electrode thickness and divided by the concentration difference. In normalized units, diffusivity and ionic conductivity are the same, as they are proportional. The diffusion simulation considers periodic boundary conditions on the external xz and yz planes, following the DEM simulations. Electronic conductivity is determined in an analogous fashion, from Ohm's law and a steady state potential distribution according to Poisson's equation. As boundary conditions, a 1 V potential difference is applied on the system along the z dimension, maintaining periodic boundary conditions on the perpendicular planes. Both calculations were performed in a normalized way, yielding dimensionless values of effective conductivities, by defining the bulk conductivity of the relevant material (SE for ionic conductivity and CBD for electronic conductivity) as 1.0.

3. Results and analysis

The slurry was determined to have an experimental density of

$$\rho_{\text{exp}} = (1.353 \pm 0.001) \text{ g cm}^{-3} \quad (3.1)$$

All force fields were calibrated in order to reach a desired density within the range of the experimental uncertainty. A final simulation density of $\rho_{\text{sim}} = (1.34 \pm 0.02) \text{ g cm}^{-3}$ was achieved. After the slurry equilibration step was completed, the force fields producing the dried electrode microstructure were re-calibrated to fit the porosity of the system. Porosity measurements on the experimental sample resulted in a porosity of

$$P_{\text{exp}} = \frac{V_{\text{pore}}}{V_{\text{electrode}}} = 0.53 \pm 0.03 \quad (3.2)$$

before calendaring. The dried, simulated electrode porosity falls into this range, achieving a value of $P_{\text{sim}} = 0.57 \pm 0.01$, which is in the error range of the experimental porosity. Afterwards, calendaring simulations, with different compression degrees were performed on the microstructure. For this proof-of-concept work, the energy was unable to reach full equilibrium. What follows is an analysis of the microstructure after calendaring. In particular, microstructural properties, as well as electronic properties of the system are presented.

A benefit of studying computational models is the immediate availability of data in comparison to experiments. Through such computa-

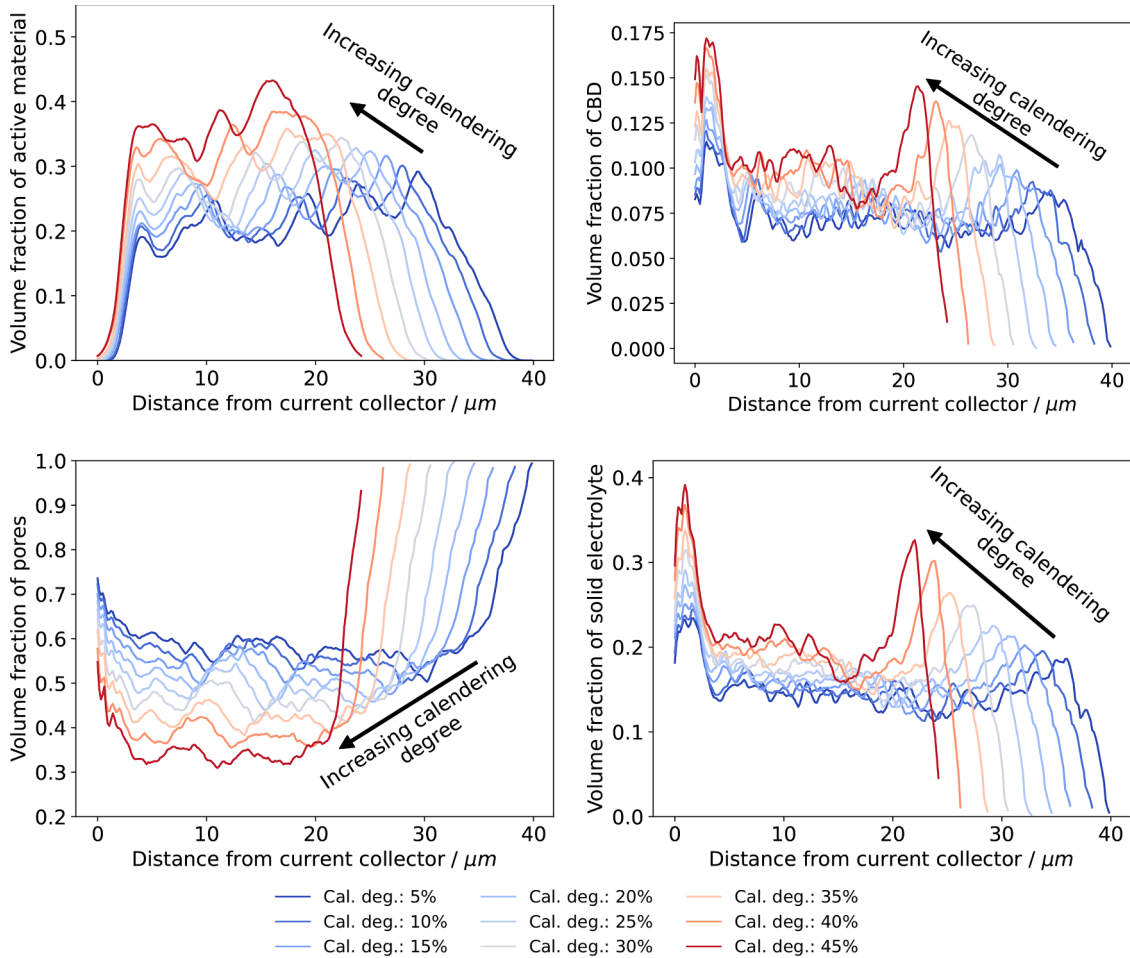


Fig. 5. Volume fraction distribution of Active Material, CBD, Pores and Solid Electrolyte Network as a function of the distance from the current collector after performing calendaring simulations to different degrees. The volume fraction is defined as the percentage of a given material within a distance $[z, z + \Delta z]$ from the current collector, relative to the volume within this volume element. All studied microstructures are in their relaxed state, after the springback has occurred. Volume fraction of active material (top left), CBD (top right), pores (bottom left) and solid electrolyte (bottom right) are shown. Different calendaring degrees are color coded.

tional models, properties of interest can be analyzed as soon as the in silico generated microstructure is complete. We therefore start with the analysis of the material distribution throughout the electrode and the change thereof upon calendaring. Volume fractions for each material phase are plotted as a function of distance from the CC in Fig. 5. A volume fraction is defined as the percentage of a given material within a distance $[z, z + \Delta z]$ from the CC, relative to the volume within this volume element. All color codes in Fig. 5 represent the different calendaring degrees of the simulations ranging from low calendaring degrees in blue to high calendaring degrees in red. Plots in Fig. 5 represent the volume fraction of active material (Fig. 5, top left), CBD (Fig. 5, top right), pores (Fig. 5, bottom left) and solid electrolyte (Fig. 5, bottom right) at a given distance from the CC. For all microstructures, which are comparable to the microstructures of electrodes obtained from experiments and dried at low drying rates, similar trends are depicted for all calendaring degrees, therefore we shall describe the effect of the distance from the calendaring degree on the volume fraction first and subsequently elucidate the impact of the calendaring degree on the volume distribution.

At short distances from the CC, the active material has little to no contribution to the total volume fraction, the value appears to be zero and only increase at several micrometers from the CC. We observe, on the other hand, a relatively large contribution of CBD and solid electrolyte to the total volume fraction, which decreases at a distance of several micrometers to the CC. We explain this behavior on account of the realistic particles shapes and relative size of the active material. Since the active material is much larger than the solid electrolyte and CBD particles, only an edge of the particle will be in contact with the CC. The center of the active material particle will be several micrometers away from the CC. Therefore, we expect only a small amount of active material close to the CC, but a gradual increase as we move away. Likewise, the space that is not filled with the active material is filled with other materials on account of the smaller size and therefore better packing. Thus explaining the larger SE and CBD volume fractions close to the CC. Similar trends are observed for all materials at intermediate distances from the CCs. All volume fractions of the materials seem to fluctuate around a constant value for intermediate distances from the CC. Such a trend is expected, because of the homogeneous distribution of the materials as well as the homogeneous drying phase of the slurry. At distances opposite to the CC, all material volume fractions decrease towards zero. Pores also appear to have a relatively constant volume fraction distribution over all distances from the CC, but on the other hand, their volume fraction increases at the top of the electrode. Both of these effects, shown on the different panels of Fig. 5 can be explained by the rugosity of the surface formed after the calendaring roll is retracted. During calendaring all materials get pushed down. Slight fluctuations are present in the volume fraction distribution on account of the slight variability in material distribution. The extent of resulting elastic recovery is therefore not homogeneous, and as a result the decrease in volume fraction occurs over the span of several micrometers when projected along the z axis. The calendaring degree appears to have a low impact on the magnitude of this rugosity.

We continue with the analysis of the porosity of the system as the calendaring degree increases. Porosities after the microstructure have relaxed as a function of calendaring degree are depicted in Fig. 6. Here, one observes that as the calendaring degree increases, the porosity of the electrode decreases steadily from approximately $P_{\text{sim}} = 0.57 \pm 0.01$ in the uncompressed state to approximately $P_{\text{sim}} = 0.30 \pm 0.02$ at a 45 % calendaring degree. The plot does not follow a linear relationship since the extent of the springback effect varies according to the applied pressure. It is evident that as the calendaring roll presses down on the dried electrode, the system under study will begin to decrease in volume on account of the added stress. This phenomenon is also reflected in the studied system, since as the pressure increases through the higher calendaring degree, the pores present within the electrode begin to shrink. As the stress is increased, first the pores begin to close, on

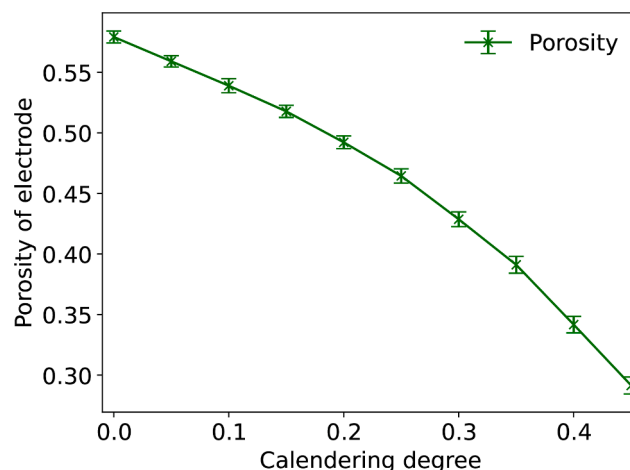


Fig. 6. Porosity of the formulation with respect to the calendaring degree. Each data point corresponds to a simulated measurement, in which the error bars represent the standard error of the data under study.

account of the high Young's modulus of the material phases present in the configuration. Such a microstructural evolution is expected and desired. During calendaring such a behavior is explicitly targeted, as it decreases the volume of inactive phases, such as pores, present in the electrode to increase the energy density of the electrode. Simultaneously the interfacial contact between all material constituents should increase, in order to allow for the formation of electronically and ionically percolation pathways, which also increases the performance of the electrode during cycling.

It is necessary to include a study of the interfacial contact area evolution between the different materials present in the cathode upon calendaring. The interfacial contacts determine the effectiveness of the intercalation/de-intercalation reactions of lithium at the interface between AM and SE. It is worth to mention that such an effectiveness might evolve upon electrochemical cycling due to the cyclic variation of AM particles volume and consequent materials interfaces delamination. In the work reported in this article, we do not simulate the electrochemical cycling, but only the interfacial contact evolution during calendaring. For better representability, we introduce the concept of a normalized interfacial area A_{norm} , which is defined as the ratio between the total interfacial area $A_{\text{interface}}$ between two distinct materials in the cathode volume and the base area of the cathode, A_{base} , in the studied ensemble. We depict the value A_{norm} for different calendaring degrees and for the six types of interfaces present between all four phases types, including pores, in our simulation in Fig. 7. We begin with the analysis of the interfacial area between the materials and pores. We observe that as the calendaring degree increases, the interfacial area between each material and pore decreases. Such a trend is in line with the observations made in Fig. 6. Namely, as the calendaring degree increases and pores begin to shrink, so does the interfacial area between material and pore. Thereby, as the pores shrink, there is less volume occupied by the pore and less possible area of contact between pore and material, thus the interfacial area between pore and material begins to decrease. The magnitude of the value of A_{norm} depends on the volume of the material present, since SE has the highest volume, it also has the largest value in the beginning. For the interfacial area between the different materials, we observe a trend which we attribute to a some of the approximations of our simulation. It appears that despite a larger calendaring degree, the interfacial area between the different materials does not seem to increase, but rather remains constant. This is counter-intuitive to our experimental understanding, because as the material is compressed more and more, we would expect particularly the binder and solid electrolyte phase to deform and increase its interfacial contact with the other materials. In our simulations, we do not observe this trend on account of the spherical

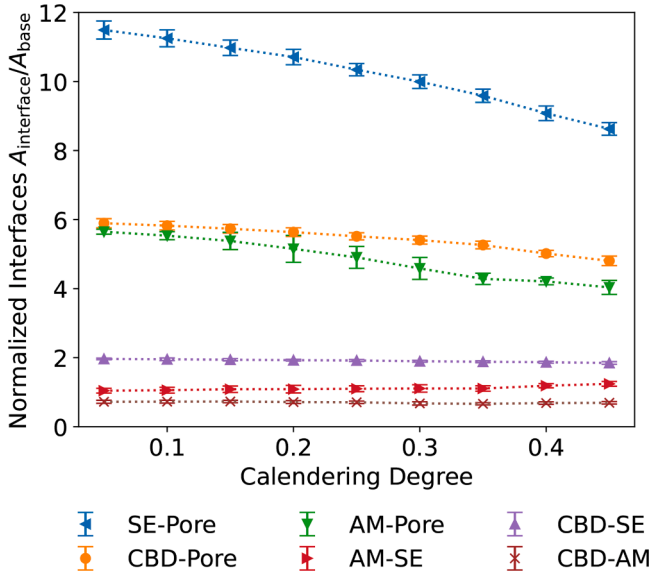


Fig. 7. Evolution of normalized interface between different material constituents present in the electrode after springback at different calendaring degrees. A normalized interfacial area is defined as the total interfacial area between two material constituents divided by the area of the studied microstructure. Error bars represent the standard error.

nature of these particles in our simulations. In essence, once contact has been established between two or more spheres, no matter how much the microstructure is compressed, two phases will not be able to increase their contact area, since we do not allow for much overlap between the different spheres during the calendaring stage in the first place.

The increased understanding of the effects of manufacturing on the microstructure, allowed by the simulation, enables us to study and explain the electronic and ionic conductivity of the system. Electronic conductivity is expected to increase as the electrode is compressed, decreasing porosity and increasing the contact between particles [50]. Similarly, unlike lithium-ion batteries, ionic conductivity is expected to increase as calendaring degree increases, due to the tighter connection between SE particles and the associated decrease in tortuosity factor of the SE phase [51,52]. Experimentally, calendaring pressure cannot be increased indefinitely though, since large pressures can induce AM particle cracking [53]. Both the electronic and ionic conductivities are shown as a function of the calendaring degree in Fig. 8.

We observe a steady increase of both the electronic and ionic conductivity as the calendaring degree is increased. This trend matches the observation of increased particle contacts in experiments [54] and in

simulations [55] and is a desired effect of increasing the calendaring degree, as this allows for better percolative pathways for both ions and electrons. It should be noted that in solid state batteries, ion transport occurs through the SE, therefore higher compression levels enhance lithium transport, as observed both in experiments [56] and in simulations [16]. The observed trends are explained with the less tortuous pathways electrons and ions need to take on account of the smaller distance between top and bottom of the microstructure, thus increasing their electronic and ionic conductivity, although we would expect even higher values for larger interfacial areas between materials. Recognizing the interest of these aspects of calendaring in optimizing manufacturing, one way to improve the realism of our computational manufacturing workflow would be to better parametrize the force field parameters used during the drying and calendaring stages according to experimental data, considering macroscopic properties such as porosity as a function of calendaring pressure, conductivities, and cracking of secondary particles.

4. Conclusions

In this article a previously developed framework for the simulation of the wet-processing steps for solid-state battery electrode manufacturing was extended to account for realistic particle shapes. Simultaneously, the model was calibrated to experimental results, so that experimental conditions can be reproduced as closely as possible. The simulation is based on an alternative to stochastic generation of microstructures, which links each step of the manufacturing process to an experimentally validated property of the electrode. In particular, in our model, we studied the effect of calendaring on the structure-property relations of the cathode. Our results matched the trends expected in experiments. In particular, we observed a homogeneous distribution of all materials throughout the microstructures. Interestingly, calendaring appears to mostly have a scaling effect on the distribution of all phases within the electrode, because all material phases are compressed along the height of the electrode and the pore shrinkage during calendaring. We expect such a behavior for slow drying rates, albeit models have been developed by us that can account for fast drying as well [29]. Qualitatively, we were able to observe the pore shrinkage in our model as well. This effect occurred non-linearly with respect to the calendaring degree for the microstructures after the springback effect. We also studied the evolution of interfaces between the different phases in our simulation. Here we observed a decrease in interfacial area for all materials and pores as we increase the calendaring degree. This effect was the most pronounced for solid electrolyte and pores, because the volume of the solid electrolyte was the highest of all the materials. As a result of some of our assumptions, we were unable to observe an increase in interfacial area between different materials on account of their spherical and rigid

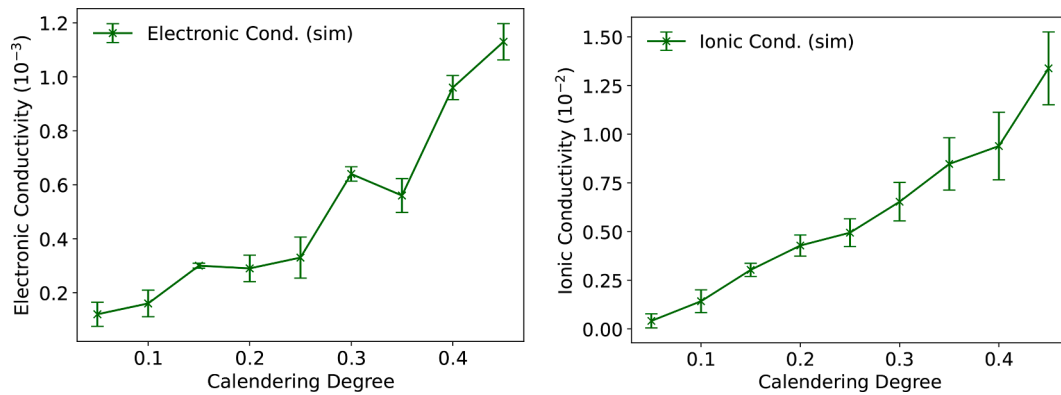


Fig. 8. Evolution of the electronic conductivity (left) and ionic conductivity (right) of the formulation shown as a function of the different calendaring degree. The conductivity values are normalized to the bulk electronic conductivity of the CBD and the ionic conductivity of the SE. All values are obtained from the structures after they have relaxed. Error bars are representative of the standard error of the values obtained from simulation repetitions.

nature. Therefore, as we increase the calendaring degree, we cannot increase the interfaces between two different materials beyond their initial contact area. Such trend is not observed experimentally and is considered as a limitation of the model. Nevertheless, we observed an increase in electronic and ionic conductivity as we increase the calendaring degree on account of the shorter percolation pathways of the charges.

Although the model is suitable for microstructure prediction, it can be improved. The porosity of an electrode is a property that is averaged over a relatively large domain. Our simulations only cover several micrometers. Using the porosity as a validation metric can lead to results that do not consider implicit structural variations within the electrode microstructure. It is however sufficient for our case, because it captures the average value of the actual microstructure. Furthermore the methodology used to simulate drying of the slurry neglects the physical intricacies encountered during the drying of actual electrodes and can only be used to capture the effects encountered in the limit of low drying rates. A setback is also the use of force field parameters that do not allow for a high degree of overlap, as this effect is important for the appropriate description of the mechanical properties of the electrode. Future iterations of this model should be calibrated by comparison with mechanical properties such as indentation curves or porosity as a function of applied pressure as our research group did in the past for Lithium Ion Battery electrodes [36].

Although these disadvantages exist, this method of microstructure generation remains one of the most complete ways of linking manufacturing with the electrochemistry of the electrode. It constitutes a good compromise between high-fidelity electrode microstructure generation and computational cost. Future iterations of such models should address different formulations to observe the effects of different microstructures on the structure-property relations of the system. It might be fruitful to study larger systems for more representability. Furthermore, these models can serve to better understand the spring-back effect observed after the calendaring rolls are removed.

CRediT authorship contribution statement

Dennis Weitz: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Franco M. Zanotto:** Writing – review & editing, Validation, Methodology, Data curation, Conceptualization. **Diana Zapata Dominguez:** Validation, Investigation, Formal analysis. **Alejandro A. Franco:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.jensm.2024.103747](https://doi.org/10.1016/j.jensm.2024.103747).

References

- [1] H. Chen, T.N. Cong, W. Yang, C. Tan, Y. Li, Y. Ding, Progress in electrical energy storage system: a critical review, *Prog. Nat. Sci.* 19 (3) (2009) 291–312. ISSN: 1002-0071
- [2] Y. Ding, Z.P. Cano, A. Yu, J. Lu, Z. Chen, Automotive Li-ion batteries: current status and future perspectives, *Electrochem. Energy Rev.* 2 (1) (2019) 1–28. ISSN: 2520-8136
- [3] M. Li, J. Lu, Z. Chen, K. Amine, 30 years of lithium-ion batteries, *Adv. Mater.* 30 (33) (2018) 1800561. ISSN: 1521-4095
- [4] P. Ping, Q. Wang, P. Huang, K. Li, J. Sun, D. Kong, C. Chen, Study of the fire behavior of high-energy lithium-ion batteries with full-scale burning test, *J. Power Sources* 285 (2015) 80–89. ISSN: 0378-7753
- [5] A. Barré, B. Deguilhem, S. Grolleau, M. Gérard, F. Suard, D. Riu, A review on lithium-ion battery ageing mechanisms and estimations for automotive applications, *J. Power Sources* 241 (2013) 680–689. ISSN: 0378-7753
- [6] K. Liu, Y. Liu, D. Lin, A. Pei, Y. Cui, Materials for lithium-ion battery safety, *Sci. Adv.* 4 (6) (2018) eaas9820. ISSN: 2375-2548
- [7] T. Ates, M. Keller, J. Kulisch, T. Adermann, S. Passerini, Development of an all-solid-state lithium battery by slurry-coating procedures using a sulfidic electrolyte, *Energy Storage Mater.* 17 (2019) 204–210. ISSN: 2405-8297
- [8] W. Zaman, K.B. Hatzell, Processing and manufacturing of next generation lithium-based all solid-state batteries, *Curr. Opin. Solid State Mater. Sci.* 26 (4) (2022) 101003. ISSN: 1359-0286
- [9] B. Paredes-Goyes, F.M. Zanotto, V. Boudeville, S. Grugeon, L. Dupont, A.A. Franco, Mesoscopic model of extrusion during solvent-free lithium-ion battery electrode manufacturing**, *Batter. Supercaps* 7 (2024) e202300441. ISSN: 2566-6223
- [10] W.B. Hawley, J. Li, Electrode manufacturing for lithium-ion batteries-analysis of current and next generation processing, *J. Energy Storage* 25 (2019) 100862. ISSN: 2352-152X
- [11] T. Inada, Fabrications and properties of composite solid-state electrolytes, *Solid State Ionics* 158 (3-4) (2003) 275–280. ISSN: 0167-2738
- [12] D. Stampatori, P.P. Raimondi, M. Noussan, Li-ion batteries: a review of a key technology for transport decarbonization, *Energies* 13 (10) (2020) 2638. ISSN: 1996-1073
- [13] R. Koerver, I. Aygün, T. Leichtweiß, C. Dietrich, W. Zhang, J.O. Binder, P. Hartmann, W.G. Zeier, J. Janek, Capacity fade in solid-state batteries: interphase formation and chemomechanical processes in nickel-rich layered oxide cathodes and lithium thiophosphate solid electrolytes, *Chem. Mater.* 29 (13) (2017) 5574–5582. ISSN: 1520-5002
- [14] G. Bucci, B. Talamini, A. Renuka Balakrishna, Y.-M. Chiang, W.C. Carter, Mechanical instability of electrode-electrolyte interfaces in solid-state batteries, *Phys. Rev. Mater.* 2 (10) (2018) 105407. ISSN: 2475-9953
- [15] A. Sakuda, A. Hayashi, M. Tatsumisago, Sulfide solid electrolyte with favorable mechanical property for all-solid-state lithium battery, *Sci. Rep.* 3 (1) (2013) 2261. ISSN: 2045-2322
- [16] M. Alabdali, F.M. Zanotto, M. Duquesnoy, A.-K. Hatz, D. Ma, J. Auvergniot, V. Viallet, V. Seznec, A.A. Franco, Three-dimensional physical modeling of the wet manufacturing process of solid-state battery electrodes, *J. Power Sources* 580 (2023) 233427.
- [17] L. Froese, J.F.v.d. Sichel, T. Loellhoeffel, L. Helmers, A. Kwade, Effect of microstructure on the ionic conductivity of an all solid-state battery electrode, *J. Electrochem. Soc.* 166 (2) (2019) A318–A328, <https://doi.org/10.1149/2.0601902jes>.
- [18] Y. Wu, S. Saxena, Y. Xing, Y. Wang, C. Li, W. Yung, M. Pecht, Analysis of manufacturing-induced defects and structural deformations in lithium-ion batteries using computed tomography, *Energies* 11 (4) (2018) 925. ISSN: 1996-1073
- [19] Z. Su, E. Decencière, T.-T. Nguyen, K. El-Amiry, V. De Andrade, A.A. Franco, A. Demortière, Artificial neural network approach for multiphase segmentation of battery electrode nano-CT images, *npj Comput. Mater.* 8 (1) (2022) 30. ISSN: 2057-3960
- [20] H. Andrä, N. Combaret, J. Dvorkin, E. Glatt, J. Han, M. Kabel, Y. Keehm, F. Krzikalla, M. Lee, C. Madonna, M. Marsh, T. Mukerji, E.H. Saenger, R. Sain, N. Saxena, S. Ricker, A. Wiegmann, X. Zhan, Digital rock physics benchmarks-part I: imaging and segmentation, *Comput. Geosci.* 50 (2013) 25–32. ISSN: 0098-3004
- [21] E.M. Ryan, P.P. Mukherjee, Mesoscale modeling in electrochemical devices-a critical perspective, *Prog. Energy Combust. Sci.* 71 (2019) 118–142. ISSN: 0360-1285
- [22] M. Alabdali, F.M. Zanotto, V. Viallet, V. Seznec, A.A. Franco, Microstructurally resolved modeling of all solid-state batteries: latest progresses, opportunities, and challenges, *Curr. Opin. Electrochem.* 36 (2022) 101127. ISSN: 2451-9103

- [23] Y. Liu, R. Zhang, J. Wang, Y. Wang, Current and future lithium-ion battery manufacturing, *iScience* 24 (4) (2021) 102332. ISSN: 2589-0042
- [24] M.M. Forouzan, C.-W. Chao, D. Bustamante, B.A. Mazzeo, D.R. Wheeler, Experiment and simulation of the fabrication process of lithium-ion battery cathodes for determining microstructure and mechanical properties, *J. Power Sources* 312 (2016) 172–183. ISSN: 0378-7753
- [25] A.C. Ngandjong, A. Rucci, M. Maiza, G. Shukla, J. Vazquez-Arenas, A.A. Franco, Multiscale simulation platform linking lithium ion battery electrode fabrication process with performance at the cell level, *J. Phys. Chem. Lett.* 8 (23) (2017) 5966–5972. ISSN: 1948-7185
- [26] F.M. Zanotto, D.Z. Dominguez, E. Ayerbe, I. Boyano, C. Burmeister, M. Duquesnoy, M. Eisentraeger, J.F. Montaño, A. Gallo-Bueno, L. Gold, F. Hall, N. Kaden, B. Muerkens, L. Otaegui, Y. Reynier, S. Stier, M. Thomitzek, A. Turetskyy, N. Vallin, J. Wessel, X. Xu, J. Abbasov, A.A. Franco, Data specifications for battery manufacturing digitalization: Current status, challenges, and opportunities, *Batter. Supercaps* 5 (9) (2022) e202200224. ISSN: 2566-6223
- [27] C. Liu, T. Lombardo, J. Xu, A.C. Ngandjong, A.A. Franco, An experimentally-validated 3D electrochemical model revealing electrode manufacturing parameters' effects on battery performance, *Energy Storage Mater.* 54 (2023) 156–163. ISSN: 2405-8297
- [28] A. Shodiev, E. Primo, O. Arcelus, M. Chouchane, M. Osenberg, A. Hilger, I. Manke, J. Li, A.A. Franco, Insight on electrolyte infiltration of lithium ion battery electrodes by means of a new three-dimensional-resolved lattice Boltzmann model, *Energy Storage Mater.* 38 (2021) 80–92. ISSN: 2405-8297
- [29] T. Lombardo, A.C. Ngandjong, A. Belhacen, A.A. Franco, Carbon-binder migration: a three-dimensional drying model for lithium-ion battery electrodes, *Energy Storage Mater.* 43 (2021) 337–347. ISSN: 2405-8297
- [30] C. Zihur, M. Lippke, A. Kwade, Model development for binder migration within lithium-ion battery electrodes during the drying process, *Batteries* 9 (9) (2023) 455. ISSN: 2313-0105
- [31] N. Susarla, S. Ahmed, D.W. Dees, Modeling and analysis of solvent removal during Li-ion battery electrode drying, *J. Power Sources* 378 (2018) 660–670. ISSN: 0378-7753
- [32] M. Nikpour, B. Liu, P. Minson, Z. Hillman, B. Mazzeo, D. Wheeler, Li-ion electrode microstructure evolution during drying and calendaring, *Batteries* 8 (9) (2022) 107. ISSN: 2313-0105
- [33] S. Wolf, M. Lippke, A. Schöo, A. Kwade, C. Schilde, A computational fluid dynamics-discrete element method model for physicsbased simulation of structure formation during battery electrode drying, *Energy Technol.* 12 (4) (2024) 2301004. ISSN: 2194-4296
- [34] C. Sangrós Giménez, B. Finke, C. Schilde, L. Froböse, A. Kwade, Numerical simulation of the behavior of lithium-ion battery electrodes during the calendaring process via the discrete element method, *Powder Technol.* 349 (2019) 1–11. ISSN: 0032-5910
- [35] A. Lundkvist, P.-L. Larsson, E. Olsson, A discrete element analysis of the mechanical behaviour of a lithium-ion battery electrode active layer, *Powder Technol.* 425 (2023) 118574. ISSN: 0032-5910
- [36] A.C. Ngandjong, T. Lombardo, E.N. Primo, M. Chouchane, A. Shodiev, O. Arcelus, A.A. Franco, Investigating electrode calendaring and its impact on electrochemical performance by means of a new discrete element method model: towards a digital twin of Li-ion battery manufacturing, *J. Power Sources* 485 (2021) 229320. ISSN: 0378-7753
- [37] J. Xu, A.C. Ngandjong, C. Liu, F.M. Zanotto, O. Arcelus, A. Demortière, A. Franco, Lithium ion battery electrode manufacturing model accounting for 3D realistic shapes of active material particles, *J. Power Sources* 9 (7) (2023) 676–682. ISSN: 1548-7105
- [38] J. Xu, B. Paredes-Goyes, Z. Su, M. Scheel, T. Weitzkamp, A. Demortière, A.A. Franco, Computational model for predicting particle fracture during electrode calendaring, *Batteries & Supercaps* 6 (12) (2023) e202300371. ISSN: 2566-6223
- [39] A. Bielefeld, D.A. Weber, J. Janek, Microstructural modeling of composite cathodes for all-solid-state batteries, *J. Phys. Chem. C* 123 (3) (2018) 1626–1634. ISSN: 1932-7455
- [40] T. Lombardo, J. Hoock, E.N. Primo, A.C. Ngandjong, M. Duquesnoy, A.A. Franco, Accelerated optimization methods for force-field parametrization in battery electrode manufacturing modeling, *Batter. Supercaps* 3 (8) (2020) 721–730. ISSN: 2566-6223
- [41] J. Schindelin, I. Arganda-Carreras, E. Frise, V. Kaynig, M. Longair, T. Pietzsch, S. Preibisch, C. Rueden, S. Saalfeld, B. Schmid, J.-Y. Tinevez, D.J. White, V. Hartenstein, K. Eliceiri, P. Tomancak, A. Cardona, Fiji: an open-source platform for biological-image analysis, *Nat. Methods* 9 (7) (2012) 676–682. ISSN: 1548-7105
- [42] C.A. Schneider, W.S. Rasband, K.W. Eliceiri, Nih image to imagej: 25 years of image analysis, *Nat. Methods* 9 (7) (2012) 671–675. ISSN: 1548-7105
- [43] I. Arganda-Carreras, V. Kaynig, C. Rueden, K.W. Eliceiri, J. Schindelin, A. Cardona, H. Sebastian Seung, Trainable weka segmentation: a machine learning tool for microscopy pixel classification, *Bioinformatics* 33 (15) (2017) 2424–2426. ISSN: 1367-4811
- [44] B.L. Trembacki, A.N. Mistry, D.R. Noble, M.E. Ferraro, P.P. Mukherjee, S. A. Roberts, Editors' choice-mesoscale analysis of conductive binder domain morphology in lithium-ion battery electrodes, *J. Electrochem. Soc.* 165 (13) (2018) E725–E736, <https://doi.org/10.1149/2.0981813jes>. ISSN: 1945-7111
- [45] A.P. Thompson, H.M. Aktulga, R. Berger, D.S. Bolintineanu, W.M. Brown, P. S. Crozier, P.J. in 't Veld, A. Kohlmeyer, S.G. Moore, T.D. Nguyen, R. Shan, M. J. Stevens, J. Tranchida, C. Trott, S.J. Plimpton, LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales, *Comput. Phys. Commun.* 271 (2022) 108171. ISSN: 0010-4655
- [46] K.L. Johnson, Contact mechanics. ISSN: 2057-3960, 1, 8, (1985).
- [47] H.A. Lorentz, Ueber die anwendung des satzes vom virial in der kinetischen theorie der gase, *Ann. Phys.* 248 (1) (1881) 127–136. ISSN: 1521-3889
- [48] T. Lombardo, F. Lambert, R. Russo, F.M. Zanotto, C. Frayret, G. Toussaint, P. Stevens, M. Becuwe, A.A. Franco, Experimentally validated three-dimensional modeling of organic-based sodium-ion battery electrode manufacturing, *Batter. Supercaps* 5 (8) (2022) e202200116, <https://doi.org/10.1002/batt.202200116>.
- [49] M. Duquesnoy, C. Liu, D.Z. Dominguez, V. Kumar, E. Ayerbe, A.A. Franco, Machine learning-assisted multi-objective optimization of battery manufacturing from synthetic data generated by physics-based simulations, *Energy Storage Mater.* 56 (2023) 50–61. ISSN: 2405-8297
- [50] H. Zheng, L. Tan, G. Liu, X. Song, V.S. Battaglia, Calendaring effects on the physical and electrochemical properties of Li[Ni1/3Mn1/3Co1/3]O2 cathode, *J. Power Sources* 208 (2012) 52–57, <https://doi.org/10.1016/j.jpowsour.2012.02.001>.
- [51] Y. Sakka, H. Yamashige, A. Watanabe, A. Takeuchi, M. Uesugi, K. Uesugi, Y. Orikasa, Pressure dependence on the three-dimensional structure of a composite electrode in an all-solid-state battery, *J. Mater. Chem. A* 10 (31) (2022) 16602–16609, <https://doi.org/10.1039/d2ta02378d>.
- [52] L. Froboese, J.F. van der Sichel, T. Loellhoeffel, L. Helmers, A. Kwade, Effect of microstructure on the ionic conductivity of an all solid-state battery electrode, *J. Electrochem. Soc.* 166 (2) (2019) A318-A328, <https://doi.org/10.1149/2.0601902jes>.
- [53] Z. Zhang, M. Bai, X. Fan, M. Yi, Y. Zhao, J. Zhang, B. Hong, Z. Zhang, G. Hu, Y. Lai, A low cost single-crystalline LiNi_{0.60}Co_{0.10}Mn_{0.30}O₂ layered cathode enables remarkable cycling performance of lithium-ion batteries at elevated temperature, *J. Power Sources* 503 (2021) 230028, <https://doi.org/10.1016/j.jpowsour.2021.230028>.
- [54] X. Lu, S.R. Daemi, A. Bertei, M.D. Kok, K.B. O'Regan, L. Rasha, J. Park, G. Hinds, E. Kendrick, D.J. Brett, P.R. Shearing, Microstructural evolution of battery electrodes during calendaring, *Joule* 4 (12) (2020) 2746–2768. ISSN: 2542-4351
- [55] M. Lippke, T. Ohnimus, F. Frankenberger, C. Schilde, A. Kwade, Drying and calendaring of lithium ion battery electrodes: a combined simulation approach, *Powder Technol.* 444 (2024) 119984, <https://doi.org/10.1016/j.powtec.2024.119984>.
- [56] H. Fathiannasab, A. Ghorbani Kashkooli, T. Li, L. Zhu, Z. Chen, Three-dimensional modeling of all-solid-state lithium-ion batteries using synchrotron transmission X-ray microscopy tomography, *J. Electrochem. Soc.* 167 (10) (2020), <https://doi.org/10.1149/1945-7111/ab9380>.